National University of computer and emerging sciences

Artificial Intelligence Lab (AI-2002)

Semester: Spring 2026
Section: BCS-6C & BCS-6J
Course Instructor: Mr. Abdullah Yaqoob

LAB # 07 - TASKS

Constraint Satisfaction Problems using OR-Tools

Submission Instructions:

- Perform all tasks. Late submissions will not be considered.
- Submit a .ipynb file with the filename formatted as: Lab 07 - Your Name - Roll Number
- Upload the file on GCR.

TASK # 01

A classroom consists of six students: Ali, Sara, Ahmed, Zoya, Bilal, and Hina—who need to be seated in a row of six seats numbered from 1 to 6. The objective is to assign each student to exactly one seat such that all seating constraints are satisfied. According to the requirements, Sara must sit next to Zoya, meaning their seat numbers must be adjacent. Ali is not allowed to sit next to Ahmed, so their assigned seats must not be consecutive. Bilal must be seated only in an even-numbered seat (2, 4, or 6), while Hina cannot be assigned seat number 1. Additionally, each seat can be occupied by only one student, and no student can occupy more than one seat. Students are required to model this problem as a Constraint Satisfaction Problem (CSP) by defining appropriate variables representing each student, specifying the domain of possible seat numbers for each variable, and formally expressing all given constraints using mathematical or logical notation.

TASK # 02

A delivery company must assign time slots to four deliveries: D1, D2, D3, and D4. The available time slots are Morning (1), Afternoon (2), and Evening (3). Each delivery must be assigned exactly one-time slot from the given options. However, certain constraints must be satisfied while assigning these slots. Delivery D1 and D2 cannot be scheduled at the same time, meaning their assigned slots must be different. Delivery D3 must be scheduled before D4, so the time slot assigned to D3 must be less than that of D4. Additionally, D2 cannot be assigned to the Morning slot (1). Students are required to model this problem as a Constraint Satisfaction Problem (CSP) by defining appropriate variables for each delivery, specifying the domain of possible time slots for each variable, and formally expressing all constraints using mathematical or logical notation, including inequality and ordering constraints.

TASK # 03

A university needs to schedule six courses: CS101, CS102, AI201, DS202, SE301, and ML401—into a limited number of time slots and rooms. There are four available time slots, represented as {1, 2, 3, 4}, and three rooms: R1 with a capacity of 40 students, R2 with a capacity of 60 students, and R3 with a capacity of 100 students. Each course has a specific number of enrolled students: CS101 has 35 students, CS102 has 50, AI201 has 45, DS202 has 60, SE301 has 30, and ML401 has 80. Some courses have overlapping student enrollments, meaning they cannot be scheduled at the same time. Specifically, CS101 conflicts with CS102, CS102 conflicts with AI201, AI201 conflicts with ML401, and DS202 conflicts with SE301.

To model this as a Constraint Satisfaction Problem (CSP), each course must be assigned both a time slot and a room. Therefore, for each course (c), two variables are defined: (timeslot[c] in {1,2,3,4}) and (room[c] in {R1, R2, R3}). The assignment must satisfy several constraints. First, conflicting courses must not be scheduled in the same time slot. Second, the assigned room for each course must have sufficient capacity to accommodate all enrolled students. Third, no two courses can be scheduled in the same room at the same time. Additionally, ML401 must be scheduled in one of the last two time slots (3 or 4), while DS202 must be scheduled in one of the morning slots (1 or 2).

Students are required to implement a function solve_course_timetabling() using the OR-Tools CP-SAT solver, where they will define the variables, encode all constraints, and compute a feasible timetable. The output should display the assigned time slot and room for each course in a clear format.

TASK # 04

A logistics company is responsible for assigning delivery tasks to a fleet of vehicles. The company has three vehicles: V1, V2, and V3—and five delivery locations labeled L1 through L5. Each delivery location must be assigned to exactly one vehicle. However, the assignment must satisfy several operational constraints. Each vehicle has a limited delivery capacity: V1 and V2 can handle at most two deliveries each, while V3 can handle up to three deliveries. Additionally, certain assignments are restricted: locations L1 and L2 cannot be assigned to the same vehicle, and location L3 must be assigned specifically to vehicle V3.

To ensure fairness in workload distribution, the difference in the number of deliveries assigned to V1 and V2 must not exceed one. Furthermore, there is a dependency constraint: if location L4 is assigned to vehicle V1, then location L5 must also be assigned to V1. This problem can be modeled as a Constraint Satisfaction Problem (CSP) where each location (l) is represented by a variable (vehicle[l] in {V1, V2, V3}). Students are required to implement a function solve_vehicle_assignment() using the OR-Tools CP-SAT solver. The solution should incorporate counting constraints for vehicle capacities, logical implication constraints for dependencies, and balancing constraints for workload distribution, and produce a valid assignment of locations to vehicles as output.

TASK # 05

A university needs to schedule final exams for five subjects: Math, Physics, AI, Operating Systems (OS), and Databases (DB). The exams are to be conducted over three days, with each day consisting of two available time slots, resulting in a total of six slots labeled from 1 to 6. Each subject must be assigned exactly one of these time slots. However, the scheduling must satisfy several constraints. Certain subjects share common students and therefore cannot be scheduled at the same time; specifically, Math cannot be scheduled with Physics, AI cannot be scheduled with OS, and OS cannot be scheduled with DB.

Additionally, no more than two exams can be scheduled on the same day. The slots are grouped into days as follows: Day 1 includes slots 1 and 2, Day 2 includes slots 3 and 4, and Day 3 includes slots 5 and 6. Students also require sufficient rest between certain exams; therefore, there must be a minimum gap of at least two slots between AI and OS. Furthermore, there is a strict ordering constraint such that the Math exam must be scheduled before the DB exam. To manage resource limitations, each slot can accommodate at most two exams.

This problem can be modeled as a Constraint Satisfaction Problem (CSP) where each subject (s) is represented by a variable (exam[s] in {1,2,3,4,5,6}). Students are required to implement a function solve_exam_schedule() using the OR-Tools CP-SAT solver. The implementation should define variables, enforce all constraints including conflict, spacing, ordering, and capacity constraints, and output a valid exam schedule assigning a slot to each subject.

THE END!